Vakint reduction benchmark

Cached RustRed versus the FORM / FMFT reduction route

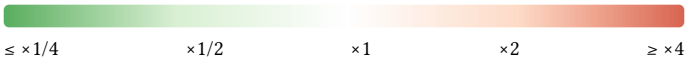
2026-10-24 · AMD EPYC 9754 128-Core Processor · CPU affinity 57

Reading guide. Bold values are RustRed times; FMFT times appear below. Multipliers are *RustRed ÷ FMFT*: green means faster, white parity, red slower. Both receive identical four-loop scalar inputs. The first physical denominator has power 2 (base) or 4 (held out); all others have power 1, auxiliaries 0.

Reduction time

Median seconds over 3 fresh processes per backend and case.

Topology	Base · power 2	Held out · power 4
H	**0.0359 s** FMFT 0.670 s ×0.054	**3.98 s** FMFT 7.37 s ×0.54
FG	**0.0450 s** FMFT 0.217 s ×0.21	**0.112 s** FMFT 0.430 s ×0.26
BMW	**0.0235 s** FMFT 0.181 s ×0.13	**0.997 s** FMFT 3.30 s ×0.30
X	**0.0822 s** FMFT 1.61 s ×0.051	**15.8 s** FMFT 21.0 s ×0.75



Method and correctness. Build profile: release (caller-declared); optimization/LTO settings are not inferred. rustc 1.98.1; FORM 5.0.0. Both retain masters in FMFT/MATAD normalization, with identical symbolic mass. Exact rational parity passed all eight cases at unit mass and renormalization scale, without truncating epsilon or using numerical tolerances.

Base cases hit prepared roots without applying rules. Held-out cases reuse cached descendants after root misses. Timing covers first public reduction calls, including FORM version checks, launch, I/O and parsing. Data loading, cache attachment, master evaluation and validation are excluded. Fresh processes run sequentially, alternating backends, with the recorded CPU affinity on a shared host. Filesystem pages are warm; this does not measure cold startup.

Selected validated run: `2026-10-24T06:59:51Z-8da99b9a5067`. Source: `8ef03d7fd755`. Data: `v0.1.2`. Samples and provenance: `data/results.json`. Refresh instructions: `../../AGENTS.md`.

Five-loop validation and benchmarks

Retained-master comparisons · functional evidence reported separately

Historical DOT ingress acceptance

2026-10-23 · one sequential debug-build pass. This checks graph definitions; it does not time IBP reduction, master evaluation, package loading or cache preparation.

Validation workload	Recorded result
Master graphs / dotted / with numerator	640 / 441 / 204
Exact routing, full numerators, native ingress and DOT round trips	Passed
Complete graph-validation pass	61.8 s
Per graph: minimum / median / maximum	0.0443 / 0.0955 / 0.175 s
Numerical master values / prepared 5L cache in this run	Absent / absent

Physical powers, signed indices, mass terms, full numerators, loop-basis slots (including zero) and integration Jacobians were checked exactly. These checks establish the stored graph definitions, without claiming that the catalogue is an independent master basis. One observed run supports no statistical speedup claim. Earlier interrupted checks are not a timing baseline.

Five-loop reduction comparison

Retained-master workload	No persistent cache	With persistent cache
Complete quadratic UV fixture	Pending	Pending
Zero-momentum diagnostic	Pending	Pending
At least 12 held-out inputs	Pending	Pending

Eligibility and reading guide. Primary cells require catalogue closure and exact result/guard agreement, a completed strict cache, and at least three matched fresh processes per mode in one accepted serialized window, with fixed workers and affinity. Bold values are reduction seconds; small values give the full observed range. Multipliers are *cached ÷ uncached*, using the same colour scale as page 1. Functional runs, including the exploratory q0 cohort, never supply these medians. Cache preparation and held-out performance retain separate evidence.

Historical ingress: source `68f2263bf54d` (modified); RustRed `52c7463686b4` Symbolica `b6cca776f9dc`. Exact patch, executable, data and per-graph digests remain in JSON.

Historical compatibility record, separate from selected performance evidence: RustRed `91b72f8f5d42` Symbolica `9c8623ecb800`. Published upgrade: Symbolica `1205ec30890f`, FeynKit `d22e0f9fdbf8` acceptance was pending in that historical record. Full revisions: `data/results.json`.

Five-loop evidence and timing ledger

Selected report: `five-loop-2f0a5a637cf939141c8b1eeb`. Raw reports and their hashed launcher/compiler receipts remain in JSON. Qualified performance additionally binds complete result/guard files, unknown-master inventories and accepted window receipts. A successful retained expression alone does not establish catalogue closure or numerical evaluability.

Primary workloads

Complete quadratic UV fixture: pending.

Zero-momentum diagnostic: pending.

Strict cache preparation

Producer `8ef03d7fd755` · original phase: `pilot`. Retained completed candidate; selection recorded separately as `828f61d0d6a6`. 2 / 2 selected roots complete; 25086 intermediate records; strict Reject policy. Compressed size: 2.342 MiB; decoded: 16593448 bytes (15.825 MiB). Original selected-pilot producer-driver elapsed: `1272.632297156` s, including loading and owner setup, not pure IBP or overall campaign time. Cache SHA-256: `cc5e6d80aacc737b81be1675a8aa02694b9ccaa1e4f31cc2d5a34199e4f1a19d`. Training roots, capture diagnostics and source/proof bindings remain in the unmodified build receipt.

Accepted native compatibility bridge `daa07861eaf7` · consumer cohort receipt `2b2a26eb74cc` · consumer workload `f022d2173668`. This does not establish a fresh target-suite pass, later-cohort compatibility or performance.

Separate rejected trial: 1 / 3 roots complete; `32613.349550541956` s whole attempt; no final gzip candidate.

Timing boundaries. Stages overlap; never sum them. Owner preparation is included in saved-application/reduction times. `ibp_excluding_owner_seconds` means saved application minus owner preparation, including transport, combination and terminal substitution. Pure IBP, terminal-only, isolated gzip decode and primary owner-subtracted times remain unmeasured. Prepared-input E2E includes current input/data/cache loading and public reduction; generator preparation stays historical. Fresh/prepared scopes never mix. Initial loading uses Sparse; representation reload belongs to reduction. Missing events are "—", never zero. Full records remain in JSON.

Genuinely connected five-loop control

Supplement: same-engine warm reuse. This scalar ladder-pinch control is separate from the physical quadratic/q0 workloads and from held-out persistent-cache comparisons. Each fresh process makes one first public API call and then repeats it on the same engine, settings and input.

1 completed functional pair(s). These observations are non-isolated; no median, range or ratio is reported. Three qualified fresh pairs in one accepted window are required for timing aggregates.

First public call · observed seconds	Repeated call · observed seconds
`21.743564929`	`6.918791091`

Functional source `10b3a9a075fa` · executable `d2cebdd00c2a` · data `4c0877b63b2d`.

Scope. Default prepared 1L–4L caches are attached; no persistent 5L cache is attached before or after either call. On-demand owner preparation is included inside the API timer. Loading, input preparation, serialization, cache snapshots and parity checks are excluded. Reuse may involve several kinds of in-memory state; this is not a selected persistent-cache speedup. Pure IBP, owner-only and terminal-only durations remain unmeasured. These values never enter the cold-process or physical-workload medians. Family identity parity establishes neither catalogue closure nor numerical master availability.